JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B3, PAGES 1481-1486, MARCH 10, 1980

Magnetic Fabric of the Elberton Granite, Northeast Georgia

BROOKS B. ELLWOOD AND JAMES A. WHITNEY

Department of Geology, University of Georgia, Athens, Georgia 30602

The anisotropy of magnetic susceptibility of samples from 21 sites within the granitic Elberton pluton, located in northeast Georgia, has been used to determine a possible tectonic rotation, source, and flow direction for granite emplacement. The data indicate that the granitic magma may have been initially emplaced in the northeastern and south central portions of the body and then flowed as a result of plastic deformation to the northeast and southwest. Estimates of tectonic rotation are based on field evidence and on the assumption that the magnetic fabric within the body was initially horizontal. Present orientation of this fabric, then, indicates a rotation about a N-NE strike, down to the southeast by possibly as much as 35°. A zone of high bulk susceptibility within the body, indicative of an excess in iron content, is closely associated with areas of pink discoloration of the granite. We suggest that the pink discoloration is the result of low-temperature oxidation of magnetite to hematite in the feldspars.

INTRODUCTION

The Elberton granite is a large, fine grained pluton widely used as monument stone and of great economic significance in northeast Georgia. It occupies an elongate northeast to southwest region in northeastern Georgia, some 60 km long and 7–10 km wide, and is bounded to the NW and SE by NE-SW trending faults (Figure 1). Locally, the granite is strongly discordant with stoped blocks of various metamorphic lithologies and older granites present.

This granite is the only major pluton cut by the recent seismic line, across the southern Appalachians, run by the Consortium for Deep Continental Reflection Profiling (Cocorp) [*Schlit et al.*, 1979; *Cook et al.*, 1979*a*, *b*]. Thus the tectonic and emplacement history of the Elberton may be of vital importance in interpreting the history of the southern Appalachians. This paper represents one phase of an extensive study of the magnetic properties of the pluton in an effort to evaluate the magmatic and tectonic history of the granite.

Magnetic susceptibility (K or χ) is a symmetrical second-rank tensor coefficient relating the two vector (first-rank tensor) properties, induced magnetic moment J_i and the inducing magnetic field H_j [*Nye*, 1969]. It is quoted in terms of the susceptibility per unit volume K, or the susceptibility per unit mass χ in SI units.

For any rock specimen an orthogonal system of maximum, intermediate, and minimum eigenvectors and eigenvalues (directions and magnitudes) of the susceptibility tensor coefficient is derived from the 3×3 susceptibility tensor matrix of direct measurements which are generally determined for a coordinate system unrelated to the coordinates defined by the principal axes. Anisotropy of magnetic susceptibility (AMS) data are conventionally expressed as triaxial ellipsoids whose major, intermediate, and minor axes represent the corresponding maximum, intermediate, and minimum susceptibility directions, with magnitudes K_a, K_b, and K_c, respectively (magnitude ellipsoid of *Nye* [1969]). Bulk susceptibility is given as

$$\bar{K} = \frac{K_a + K_b + K_c}{3} \quad (1)$$

For igneous rock specimens it has been experimentally and theoretically shown that the total susceptibility ellipsoid shape expresses the preferred alignment of ferrimagnetic mineral grains [e.g., *Khan*, 1962].

Graham [1954] suggested that AMS measurements may be a potentially valuable method for evaluating petrologic relationships in rocks. *King* [1966] followed up on this comment and on the work of *Khan* [1962], which dealt with metamorphic and basic igneous rocks, in his AMS evaluation of some Irish granites. He touched on several important points:

1. The presence of magnetite, even in small amounts, overshadows the magnetic effects of hematite. The AMS in such samples in contributed by the magnetite grains.
2. The measured AMS reflects the orientation of the magnetite grains and therefore results from petrologic factors affecting grain orientations.
3. Secondary magnetite produced within ferromagnesian minerals (for example, mica) records the primary fabric of the host grain. *Birch* [1979] has made similar observations for the Exeter pluton.
4. A marked foliation parallel to the granitic contact with the country rock is exhibited, becoming more intense as the contact is approached. This was also shown by *Van der Voo and Klootwijk* [1972] and *Birch* [1979].
5. Plastic flow is recorded by a long-axis AMS alignment parallel to the flow direction.
6. Good groupings of short AMS directions result from compressive stresses and are parallel to such stress directions.

Other AMS work on granites has centered around the orientation of the magnetic fabric [*Balsley and Buddington*, 1960], or how that fabric relates to remanent magnetism (RM) acquisition. *Heller* [1973], for example, concluded that the magnetite in the Bergell Massif (Switzerland) is a secondary mineral which grew in response to stress after the granite was emplaced and therefore the RM carrier is secondary. Other studies in granites have addressed the problem of the deflection of an RM direction as a result of high percentage of anisotropy [e.g., *Van der Voo and Klootwijk*, 1972].

METHODS AND RESULTS

Field. A total of 136 cores were taken with a gasoline-powered portable drill from 20 quarries and one blasted road cut in the Elberton Pluton (Figure 1) and oriented using a Brunton compass. Core sites were selected which avoided (1) rocks within 50 cm of flame cut surfaces, (2) weathered surfaces, and (3) any unusual features such as xenoliths, pegma-

Copyright © 1980 by the American Geophysical Union.

Paper number 9B1603.
0148-0227/80/009B-1603$01.00

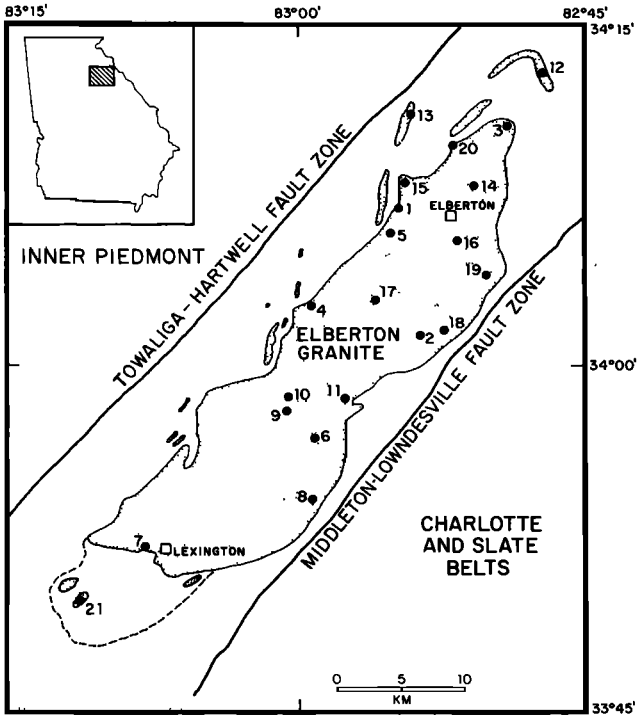


Fig. 1. Location of the Elberton granite (stippled), Georgia, in relationship to the Inner Piedmont, Charlotte and Slate belts, Towaliga-Hartwell Fault, and the Middleton-Lowndesville Fault. Diagonals are shear zones. Dashed line delineates the Hutchins Granite, which lies in close association to the Elberton. Sample sites are numbered.

tites, or sheared surfaces. Cores were separated laterally by an average distance of greater than 5 m.

Laboratory. The cores, of ~2.5-cm diameter, were sliced into lengths of ~2.2 cm to conform to an optimum length to diameter ratio range of 0.82–0.90 [*Banerjee and Stacey*, 1967; *Noltimier*, 1971]. The AMS was measured for six samples from each site using an automated low-field torque magnetometer for which the precision has been determined [*Ellwood*, 1978*a*]. Calibration entailed measuring a machined, seamless copper ring [*Noltimier*, 1964]. Sixteen samples were selected from several sites, and polished sections were prepared and examined under reflected light.

Statistical. AMS is a symmetric physical property with the induced moment always in the same sense as the applied or inducing field. Thus a principal AMS axis defines two directions on the unit sphere, and a random scatter of the axial directions about some preferred azimuth defines a bimodal distribution of induced moment orientations, quite unlike the unimodal scatter of remanent magnetic vectors. Therefore principal axis mean directions were calculated using the eigenvalue method, summarized by *Mark* [1973], which is independent of any one direction assigned to a principal axis.

Within-site scatter about the AMS axial mean is represented here by

$$S_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \tag{2}$$

and λ_1, λ_2, and λ_3 are the eigenvalues of the mean AMS azimuth. S_1 can range from 1.0, representing a cluster of identical axial orientations and a unidimensional representation quadric, to 0.333 for a random scatter of orientations whose representation quadric is a sphere.

Numerical. A useful parameter employed in this study is percent of anisotropy, where

$$\% \text{ An} = \frac{K_a - K_c}{K_b} \times 100 \tag{3}$$

TABLE 1. Anisotropy of Magnetic Susceptibility Means

	K_a			K_c							
N	D, deg	I, deg	S_1	D, deg	I, deg	S_1	α_{95}	% An	$\bar{K} \times 10^{-4}$	F/L	E
1	357.2	−0.3	0.603	272.2	57.1	0.942	12.9	2.96 ± 0.49	2.41 ± 0.06	5.52 ± 5.85	1.016 ± 0.005
2	188.1	−1.0	0.872	271.9	76.3	0.777	28.5	4.09 ± 2.14	29.30 ± 0.06	2.07 ± 1.50	1.014 ± 0.022
3	347.9	−25.2	0.963	284.8	42.6	0.964	10.0	5.30 ± 0.57	4.49 ± 0.09	2.49 ± 1.09	1.021 ± 0.077
4	211.2	−24.9	0.710	283.5	32.6	0.917	15.7	4.71 ± 1.69	5.32 ± 0.14	2.43 ± 1.43	1.016 ± 0.010
5	199.5	−16.0	0.802	250.2	61.6	0.941	13.0	8.48 ± 6.39	3.77 ± 0.19	1.05 ± 0.41	1.010 ± 0.018
6	208.8	−22.0	0.944	279.8	41.0	0.982	7.1	4.36 ± 2.32	3.46 ± 0.08	2.69 ± 2.84	1.016 ± 0.028
7	339.8	−20.9	0.809	298.5	63.2	0.969	9.3	2.99 ± 0.53	4.81 ± 0.15	2.20 ± 2.12	1.007 ± 0.007
8	286.5	−28.0	0.643	294.0	65.3	0.973	8.7	2.40 ± 0.60	2.69 ± 0.05	3.71 ± 0.95	1.014 ± 0.004
9	338.6	−15.4	0.928	294.1	69.9	0.937	13.4	2.30 ± 0.78	5.13 ± 0.09	2.18 ± 1.58	1.006 ± 0.009
10	322.6	−21.5	0.790	286.1	64.3	0.956	11.0	5.28 ± 1.91	19.58 ± 0.06	6.86 ± 7.66	1.031 ± 0.014
11	219.8	−22.9	0.813	297.7	42.9	0.881	19.0	1.66 ± 1.23	5.90 ± 0.19	1.15 ± 0.74	1.000 ± 0.002
12	348.1	−27.9	0.866	302.2	56.7	0.938	13.3	5.99 ± 5.75	3.38 ± 0.15	5.18 ± 9.10	1.022 ± 0.047
13	330.7	−22.8	0.967	301.7	63.8	0.993	4.4	3.73 ± 1.04	4.51 ± 0.20	3.58 ± 1.48	1.020 ± 0.007
14	322.4	−1.3	0.939	236.4	81.1	0.917	15.5	3.30 ± 0.91	5.53 ± 0.12	0.88 ± 0.42	0.997 ± 0.008
15	345.1	−9.8	0.973	267.3	50.1	0.992	4.7	6.73 ± 8.67	3.95 ± 0.19	3.03 ± 1.73	1.041 ± 0.068
16	343.7	−6.5	0.641	261.1	55.6	0.907	16.8	2.05 ± 0.65	6.30 ± 0.15	1.70 ± 1.20	1.002 ± 0.006
17	357.0	−2.9	0.915	270.0	51.0	0.928	14.4	1.82 ± 0.52	6.17 ± 0.09	1.39 ± 0.74	1.003 ± 0.006
18	243.5	−20.3	0.754	299.6	50.8	0.909	16.4	3.01 ± 1.08	16.65 ± 0.04	1.92 ± 1.07	1.007 ± 0.003
19	3.5	−17.2	0.841	290.0	28.7	0.772	28.0	4.44 ± 2.03	19.92 ± 0.05	0.42 ± 0.21	0.983 ± 0.009
20	303.7	−18.7	0.811	281.0	74.8	0.895	17.7	1.20 ± 0.59	3.55 ± 0.06	1.20 ± 0.59	1.007 ± 0.005
21	335.3	−5.1	0.597	262.7	57.6	0.970	9.1	10.29 ± 1.11	19.19 ± 0.03	7.62 ± 4.21	1.081 ± 0.021
Mean	348.3	−8.9	0.722	282.1	57.6	0.927	···	4.15 ± 2.30	8.38 ± 7.58	2.82 ± 1.97	1.015 ± 0.019

N is site number, six samples per site; k_a is AMS long axis, K_c is AMS short axis; $\bar{D}$ is mean declination, $\bar{I}$ is mean inclination where azimuths are transposed into the upper hemisphere and lower hemisphere for K_a and K_c means, respectively; % An from (3); $\bar{K}$ from equation (1), in SI units; F/L is *King*'s [1966] magnitude of magnetic fabric indicator (equation (5)); E from (4). K_a and K_c means were determined using the eigenvalve method summarized by *Mark* [1973]. S_1 is principal axis cluster indicator ((2), see text); α_{95} is after *Fisher* [1953], given for comparison, and identical with θ from the vectorial rotation method of *Mark* [1971].

21562202b, 1980, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB03p01481 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

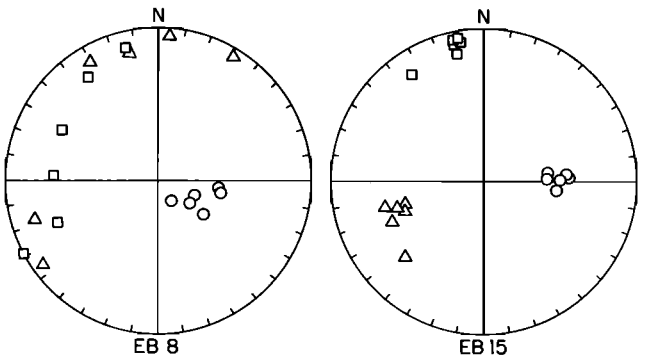


Fig. 2. Equal-area upper hemisphere projections of typical site (EB 8 and EB 15) data. Squares are K_a; triangles are K_b; circles are K_c.

and yields estimates of the magnitude of the magnetic fabric. *Hrouda and Janak* [1971] have used a parameter E, such that

$$E = \frac{K_b^2}{K_a K_c} \quad (4)$$

represents departures in shape of the AMS ellipsoid from a sphere. If E is greater than 1.0, the shape is oblate (flattened) and is indicative of a magnetic foliation for the sample. When E is less than 1.0, magnetic lineation is dominant, and the shape is prolate (elongated). *King* [1966] has quantified the magnetic fabric using the expression

$$F/L = (K_b - K_c)/(K_a - K_b) \quad (5)$$

to emphasize strong foliated fabric (large F/L) or strong lineation (small F/L).

The K_a and K_b AMS principal axes for a single site are used here to represent the within-site maximum susceptibility plane. For sites which have mainly oblate samples (mean $E >$ 1.0) and clustered K_c axes, this maximum susceptibility plane

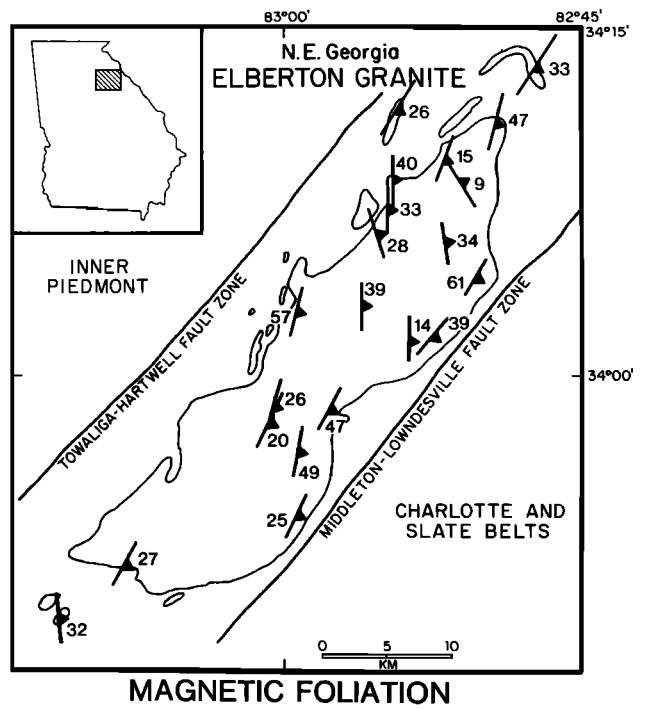


Fig. 3. Magnetic foliation for all sites within the Elberton Pluton. Data are from Table 1. Dip values are $90 - K_c I^0$ (see text) and are rounded off for clarity.

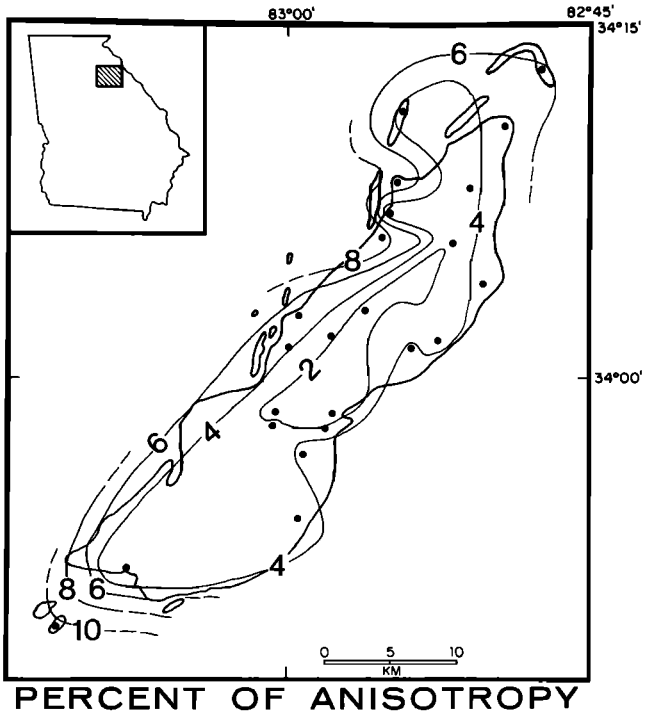


Fig. 4. Percent anistropy contours for the Elberton. Data are from Table 1.

represents the orientation of the magnetic foliation for the site. This is easiest to quantify by first determining the mean K_c azimuth which lies at the pole of the magnetic foliation plane. The strike of the foliation plane is a horizontal direction within the K_a and K_b plane and normal to the K_c mean. The dip of the plane is equal to 90 minus the absolute value of the mean K_c inclination. The magnetic lineation direction lies within this plane. We use S_1 as an indicator of the precision of the K_c axial cluster and to establish the confidence limits for the mean K_a axis for each site, indicative of the direction of magnetic lineation. Also given for comparison are the α_{95} [*Fisher*, 1953] for K_c clusters.

Mean data from the AMS measurements for each site are reported in Table 1. Also reported are S_1 as an indicator of the

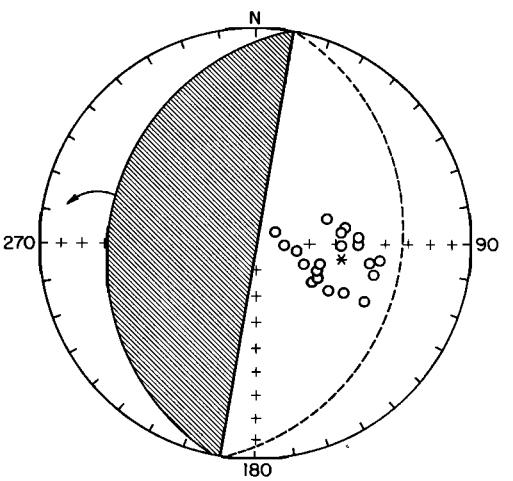


Fig. 5. Equal-area upper hemisphere projection of K_c (open symbols) site mean values. Shaded plane is the upper projection of the Elberton magnetic foliation plane. The asterisk is the K_c mean for all sites. Arrow denotes direction necessary to bring the magnetic foliation plane into a horizontal orientation.

21562202b, 1980, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB03p01481 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

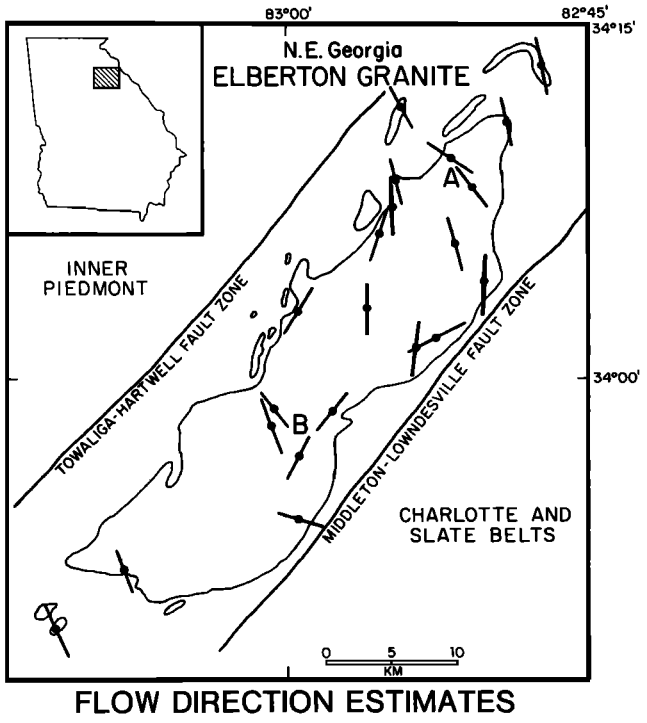


Fig. 6. Flow direction based on AMS long axis orientation (bars). A and B are inferred possible magma sources.

precision of each grouping. Examples of the types of data distributions within a single site are given in Figure 2.

Optical. Examination of polished sections indicates that magnetite is present in relative abundance. It is apparently premagmatic or very early postmagmatic and is responsible for the magnetic fabric observed. Also present in relative abundance are ilmenite grains containing exsolution lamellae of hematite. Thin sections of pink granite samples show the blood red coloration characteristic of very fine grained hematite.

DISCUSSION

Magnetic fabric. The magnitude of magnetic fabric yields oblate shapes ($E > 1.0$) and a strong foliation ($F/L > 1.0$) for all sites except EB 14 and EB 19 (Table 1), where prolate shapes are exhibited. AMS patterns for the 21 Elberton sites yield three typical types of within-site data distributions. For most sites, orientations yield K_a and K_b great circle distributions with near-vertical K_c axial clusters (for example, Figure 2, EB 8). Two sites, EB 13 and EB 15, yield distinct axial clusters for all AMS principal axes (exemplified by Figure 2, EB 15), and two sites exhibit indistinct principal axial clusters (EB 2 and EB 19). The orientation of magnetic foliation defined by within-site AMS distributions is given in Figure 3. Precision for the numbers is represented by K_c S_1 (Table 1). While a distinct macroscopic foliation exists in the Elberton near the margins of the pluton, it disappears rapidly just a few meters into the body. Where this foliation is observed, however, it mirrors the magnetic foliation.

Trends in the magnetic foliation are similar throughout the pluton, indicating an extreme uniformity which is not expected for a granite as large as the Elberton (>600-km^2 outcrop). Further, when the magnitude of anisotropy (% an from (3)) is plotted (Figure 4), it becomes apparent that magnitudes are lowest away form body margins. This is consistent with the results of *King* [1966]. Field mapping along the margins and across the northeastern section of the body yields an undulatory pattern of granite-country rock exposure, due to the nonuniform upper surface of the pluton, which, we infer, has only been partly exposed.

Tectonic rotation. Field and AMS data are consistent with the hypothesis that the present exposure of the Elberton Pluton represents the original top of the body. If this is true, it should follow that the near-margin foliation in the body will mirror the boundary orientation: since the upper boundary is subhorizontal, the observed magnetic foliation is expected to be nearly horizontal. For this to be the case it is necessary that the site AMS inclinations of each magnetic foliation plane is similar in magnitude and direction over the entire extent of the body. Systematic variability from one side to the other would indicate a change in orientation of the contact, but such a change is not observed. Magnetic foliation plane inclinations are not horizontal, however, but dip to the east by 32.4°, on average, about a strike of 12.1° (Figure 3; Table 1), possibly indicative of a physical rotation of the body at some time after emplacement. Further, sketchy field evidence in conjunction with published studies showing uplift of the Inner Piedmont [*Dallmeyer*, 1978] indicates that the Elberton pluton has experienced a slight amount of tectonic rotation about a approximate NE-SW strike. From these observations we infer that the Elberton Pluton may have undergone a slight amount of tilting about a N to NE strike, having been rotated down to the east by possibly as much as 35°. Using the site mean K_c values, we can suggest a structural reorientation for the unit (represented in Figure 5).

Emplacement. King [1966] has suggested that AMS K_a alignment results from flow. When the magma behaves as a viscous fluid, as is the case for basalitc magmas, a magnetic foliation is developed parallel to flow planes with K_a axes contained within the plane aligned normal to the flow direction. Such a result has been reported by *Khan* [1962] for basaltic lavas and dikes and by *Ellwood* [1978*b*] for Icelandic dikes.

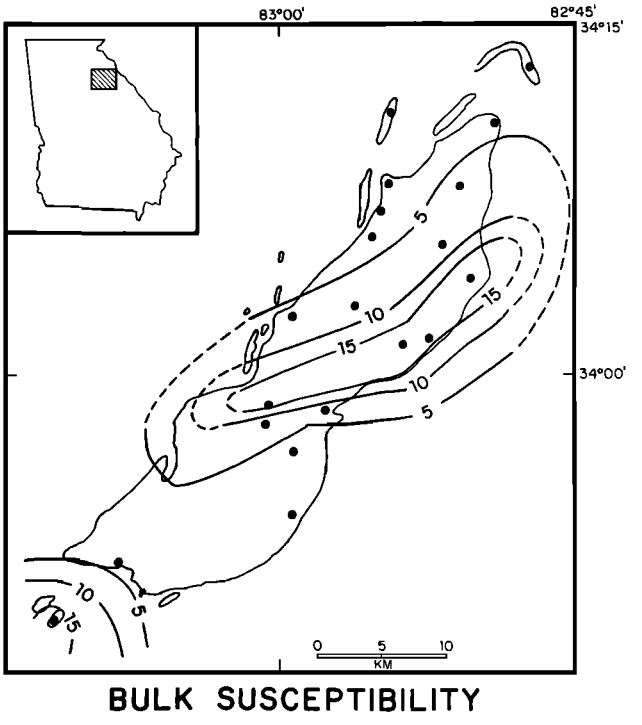


Fig. 7. Bulk susceptibility ($\bar{K}$) variability within the Elberton Pluton.

21562202b, 1980, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB03p01491 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

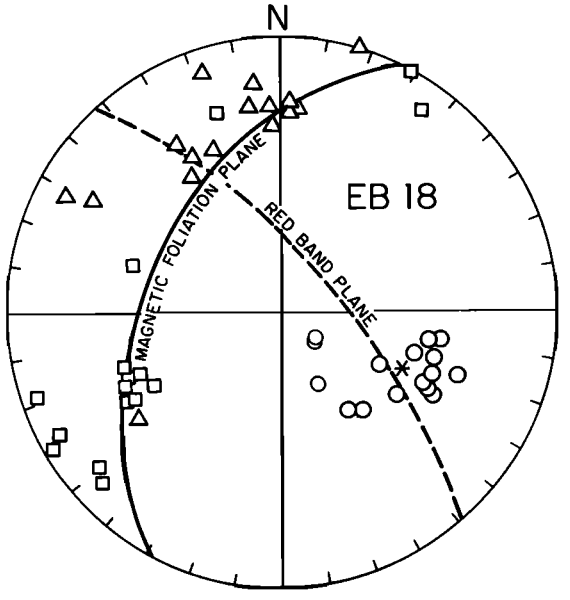


Fig. 8. Equal-area upper hemisphere projection for 15 samples from stie EB 18. Magnetic foliation plane is defined in text. The asterisk is the mean K_c direction. Red band plane orientation (dashed great circle) was determined from field measurements.

However, *King* [1966] suggests that deep-seated intrusives (granitic rocks), during the later stages of flow, behave as plastic fluids and rigid particle (magnetic grain) alignments result from plastic strain, with long axes of such grains oriented parallel with the direction of maximum strain (the plastic flow direction). Therefore plastic flow or deformation during emplacement of granites results in magnetic grain long-axis clusters parallel to the flow direction. Mean K_a azimuths are plotted in Figure 6. These orientations, representing plastic flow directions, appear to radiate outward from a central area in the northeastern (A, Figure 6) and south central (B, Figure 6) portions of the pluton. We suggest that these orientations may represent outward flow of the source magma from sites located in these areas of the body. Ideally, we need many more sites in the pluton to quantify these observations adequately, but such sites are very limited, since the body is not heavily quarried elsewhere. However, it does appear that the Elberton was emplaced and flowed northeast and southwest from at least one source and possibly several sources.

Plastic flow directed along these azimuths may help explain some of the unusual color characteristics within the Elberton. While most of the Elberton is relatively uniform grey in color, a few quarries (part of EB 2 and EB 18, and EB 19) are uncharacteristically pink. There are no apparent differences in the optical or chemical properties between the pink and grey phases. Bulk magnetic susceptibility $\bar{K}$, however, is higher in these quarries (Table 1), indicating a northeast-southwest oriented trend of higher magnetite content (Figure 7), since it has been shown for low $\bar{K}$ that $\bar{K}$ is approximately linearly related to volume percent magnetite content (discussed, for example, by *Nettleton* [1976]). This relationship indicates an approximate order of magnitude difference in magnetite content from a high in EB 2 to a low at site EB 1 and corresponds to an upper magnetite content of approximately 0.1% and a lower limit of approximately 0.01%. EB 10 and EB 21 also have high $\bar{K}$ values, but while EB 10 is consistent with the high $\bar{K}$ trend across the body (Figure 7), it is not associated with a pink phase. EB 21, also with no apparent pink phase, is isolated from the rest of the body, and no trends for this area are definitive. We infer that this relatively high magnetite content is indicative of higher iron content in that portion of the melt. Since hematite is apparently locally abundant in this zone, we suggest that low-temperature oxidation of fine grained magnetite appears to have produced very fine hematite grains within the feldspars and therefore may be responsible for the pink coloration.

In quarry site EB 18 there are 10- to 30-cm bands of color, alternating between pink and grey. The orientation of the plane of the pink bands is illustrated in Figure 8 (red band plane). This plane is at nearly right angles to magnetic foliation plane, the trend of the Elberton Pluton in outcrop pattern, and the suggested plastic flow direction for the site (represented by the mean K_a direction). One possible explanation for this pink coloration trend normal to all other trends in the area is that during emplacement, flow jointing occurred as a tensional plastic flow or deformational feature. These zones of weakness in the magma then provided conduits for the flow of volatites, and lower-temperature oxidation occurred and produced the very fine grained hematite phase which has locally colored the granite pink. These joints were later annealed as the body ceased moving and cooled, since no evidence of such features exist.

SUMMARY

On the basis of anisotropy of magnetic susceptibility analyses of 136 samples from 21 sites within the Elberton Pluton, located in northeast Georgia, we have concluded the following:

1. Sources for granite intrusion in the northeastern and south central portions of the body are suggested, as is a northeast-southwest directed emplacement flow direction, crystal orientations resulting from plastic deformation.
2. It is assumed that the magnetic foliation in the body was initially horizontal, then the pluton rotated downward by possibly as much as 35° to the southeast about a N-NE strike after emplacement.
3. The pink coloration (the only major macroscopic petrologic difference over the extent of the body, with the exception of a biotite schlieren in one quarry) is associated with higher susceptibilities and is probably due to a local increase in iron content and to local low-temperature oxidation of magnetite to hematite in the feldspar.
4. Pink bands developed at site 18 may represent annealed terminal joints which were the result of flow.

Acknowledgments. We wish to thank the granite companies in and around Elberton, Georgia, for allowing us free access to their quarries for purposes of sample acquisition. We are especially grateful to the Commolli and Coggins Granite Companies, which allowed us to return to their quarries several times over a period of months and were most helpful. Thanks are also expressed to the many graduate students who helped us with the field work, especially William Avera and Jim Hess. This research was supported by National Science Foundation grant EAR79-19911.

REFERENCES

Balsley, J. R., and A. F. Buddington, Magnetic susceptibility anisotropy and fabric of some Adirondack granites and orthogneisses, *Amer. J. Sci., 258-A*, 6–20, 1960.

Banerjee, S. K., and F. D. Stacey, The high field torque-meter method of measuring magnetic anisotropy of rocks, in *Methods in Paleomagnetism*, edited by D. W. Collinson et al., pp. 470–476, Elsevier, New York, 1967.

21562202b, 1980, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB03p01481 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Birch, F. S., Magnetic fabric of the Exeter pluton, New Hampshire, *J. Geophys. Res., 84,* 1129–1137, 1979.

Cook, F. A., D. S. Albaugh, L. D. Brown, S. Kaufman, and J. E. Oliver, Preliminary interpretation of Cocorp seismic reflection data across the Brevard zone in Northeast Georgia, paper presented at the 28th Annual Meeting, S. E. Section, Geol. Soc. of Amer., Blacksburg, Va., 1979*a*.

Cook, F. A., D. S. Albaugh, L. D. Brown, R. D. Hatcher, S. Kaufman, and J. E. Oliver, Preliminary interpretation of Cocorp reflection profiles across the Brevard zone in northeast Georgia (abstract), *Eos Trans. Agu, 60,* 314, 1979*b*.

Dallmeyer, R. D., $^{40}Ar/^{39}Ar$ incremental release ages of hornblende and biotite across the Georgia Inner Piedmont: Their bearing on Late Paleozoic–Early Mesozoic tectonothermal history, *Amer. J. Sci., 278,* 124–149, 1978.

Ellwood, B. B., Measurement of anisotropy of magnetic susceptibility: A comparison of the precision of torque versus spinner magnetometer systems, *J. Phys. E, 11,* 71–75, 1978*a*.

Ellwood, B. B., Flow and emplacement direction determined for selected basaltic bodies using magnetic susceptibility anisotropy measurements, *Earth Planet. Sci. Lett., 41,* 254–264, 1978*b*.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc., Ser. A, 217,* 295–305, 1953.

Graham, J. W., Magnetic susceptibility anisotropy, an unexploited petrofabric element, *Geol. Soc. Amer. Abstr. Programs, 65,* 1257, 1954.

Heller, F., Magnetic anisotropy of granitic rocks of the Bergell Massif (Switzerland), *Earth Planet. Sci. Lett., 20,* 180–188, 1973.

Hrouda, F., and F. Janak, A study of the mimetic fabric of some red sediments on the basis of their magnetic susceptibility anisotropy, *Sed. Geol., 6,* 187–199, 1971.

Khan, M. A., The anisotropy of magnetic susceptibility of some igneous and metamorphic rocks, *J. Geophys. Res., 67,* 2873–2885, 1962.

King, R. F., The magnetic fabric of some Irish granites, *Geol. J., 5,* 43–66, 1966.

Mark, D. M., Rotational vector procedure for the analysis of till fabrics, *Geol. Soc. Amer. Bull., 82,* 2661–2665, 1971.

Mark, D. M., Analysis of axial orientation data including till fabrics, *Geol. Soc. Amer. Bull., 84,* 1369–1374, 1973.

Nettleton, L. L., Elementary gravity and magnetics for geologists and seismologists, *Monogr. 1,* 121 pp., Soc. of Explor. Geophys., Tulsa, Okla., 1976.

Noltimier, H. C., Calibration of a spinner magnetometer with a wire loop, *J. Sci. Instrum., 41,*55, 1964.

Noltimier, H. C., A model for grain dispersion and magnetic anisotropy in sedimentary rocks, *J. Geophys. Res., 76,* 3990–4002, 1971.

Nye, J. F., *Physical Properties of Crystals,* 322 pp., Oxford University Press, New York, 1969.

Schilt, S., J. Oliver, L. Brown, S. Kaufman, D. Albaugh, J. Brewer, F. Cook, L. Jensen, P. Krumhansl, G. Long, and D. Steiner, The heterogeneity of the continental crust: Results from deep crustal seismic reflection profiling using the vibroseis technique, *Rev. Geophys. Space Phys., 17,* 354–368, 1979.

Van der Voo, R., and C. T. Klootwijk, Paleomagnetic reconnaissance study of the Flamanville granite, with special reference to the anisotropy of its susceptibility, *Geol. Mijnbouw, 51,* 609–617, 1972.

(Received June 14, 1979;
accepted October 4, 1979.)

21562202b, 1980, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB03p01481 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License